

Dynamic Programming

Chapter 6: Stochastic Discounting

Thomas J. Sargent and John Stachurski

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Topics

- State-dependent discounting: motivation
- State-dependent discounting: valuation
- Asset pricing
- MDPs with state-dependent discounting

Motivation

One limitation of MDP model: discount rate is constant

This assumption can be problematic

Example. Cannot handle time preference shocks

- Krusell-Smith 1998
- Woodford 2011
- Christiano et al. 2011, 2014
- Schorfheide et al. 2018
- etc, etc.

Also, firms discount future profits using interest rates — which are stochastic and time-varying

Example.

- nominal rates for safe assets like US T-bills
- real interest rates
- rental cost of capital

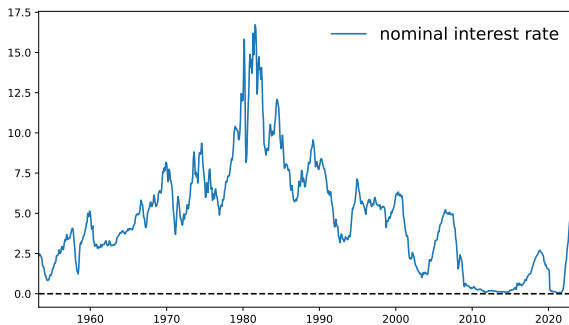


Figure: Nominal US interest rates

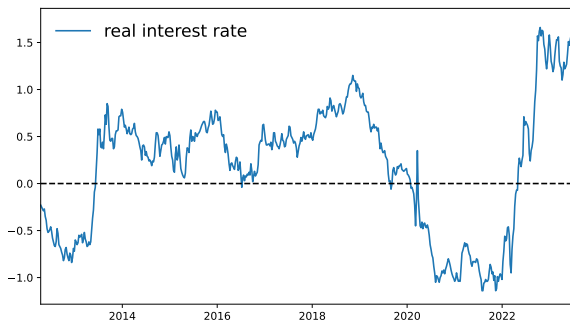


Figure: Real US interest rates

In this chapter we extend the MDP model to handle state-dependent discounting

Learn how to

1. compute lifetime values under state-dependent discounting
2. maximize these values via dynamic programming

We start with step 1...

Valuation under state-dependent discounting

Example. Consider a firm valuation problem where the interest rate follows stochastic process $(r_t)_{t \geq 0}$

The time zero expected present value of time t profit π_t is

$$\mathbb{E} \{ \beta_1 \cdots \beta_t \cdot \pi_t \} \quad \text{where} \quad \beta_t := \frac{1}{1 + r_t}$$

The expected present value of the firm is

$$V_0 = \mathbb{E} \sum_{t=0}^{\infty} \left[\prod_{i=0}^t \beta_i \right] \pi_t \quad \text{where} \quad \beta_0 := 1$$

Questions:

- When is this valuation finite?
- How can we compute it?
- Are there any general results?

The next section answers these questions

Generalized geometric sums

Suppose

- X is finite and $P \in \mathcal{M}(\mathbb{R}^X)$
- $h \in \mathbb{R}^X$
- b is a map from $X \times X$ to $(0, \infty)$

Let $(X_t)_{t \geq 0}$ be P -Markov and let

$$\beta_t := b(X_{t-1}, X_t) \text{ for } t \in \mathbb{N} \quad \text{with} \quad \beta_0 := 1$$

Let L be the **discount operator** defined by

$$L(x, x') := b(x, x')P(x, x')$$

Theorem. If $\rho(L) < 1$, then, for all $x \in X$,

$$v(x) := \mathbb{E}_x \sum_{t=0}^{\infty} \left[\prod_{i=0}^t \beta_i \right] h(X_t) < \infty$$

Moreover, this function v satisfies

$$v = (I - L)^{-1}h = \sum_{t \geq 0} L^t h$$

Remark: $I - L$ is bijective by the Neumann series lemma (NSL)

Proof: Let all the primitives be as stated with $\rho(L) < 1$

As a first step, note that, for all $t \in \mathbb{N}$, $h \in \mathbb{R}^X$ and $x \in X$,

$$\mathbb{E}_x \left[\prod_{i=0}^t \beta_i \right] h(X_t) = (L^t h)(x)$$

Proof for $t = 1$:

$$\mathbb{E}_x \beta_1 h(X_1) = \sum_{x'} h(x') b(x, x') P(x, x') = (Lh)(x)$$

Ex. Extend this argument to general t

- Hint: use induction and law of iterated expectations
- A solution can be found in the book (Ch. 6)

Interpretation:

$(L^t h)(x) =$ time zero present value of $h(X_t)$ given $X_0 = x$

Now, fixing $x \in X$, we have

$$\begin{aligned} v(x) &= \mathbb{E}_x \sum_{t=0}^{\infty} \left[\prod_{i=0}^t \beta_i \right] h(X_t) \\ &= \sum_{t=0}^{\infty} \mathbb{E}_x \left[\prod_{i=0}^t \beta_i \right] h(X_t) \\ &= \sum_{t=0}^{\infty} (L^t h)(x) \end{aligned}$$

Pointwise, this is $v = \sum_{t \geq 0} L^t h$

By the NSL and $\rho(L) < 1$, we have $\sum_{t \geq 0} L^t h = (I - L)^{-1} h$

Example. Consider the simple (constant discount case)

$$b \equiv \beta \in (0, 1)$$

Then

- $\prod_{i=0}^t \beta_i = \beta^t$
- $L = \beta P$
- $\rho(L) = \rho(\beta P) = \beta < 1$
- $v = (I - L)^{-1}h = (I - \beta P)^{-1}h$

Example. Consider again the firm valuation problem with

$$v(x) = \mathbb{E}_x \sum_{t=0}^{\infty} \left[\prod_{i=0}^t \beta_i \right] \pi_t \quad \text{where} \quad \beta_0 := 1$$

Suppose

- (X_t) is P -Markov on X
- $\beta_t = \beta(X_t)$ for some fixed $\beta \in \mathbb{R}^X$
- $\pi_t = \pi(X_t)$ for some fixed $\pi \in \mathbb{R}^X$

Let

$$L(x, x') := \beta(x)P(x, x')$$

Ex. Show the following: If $\rho(L) < 1$, then v is finite and satisfies

$$v(x) = \mathbb{E}_x \sum_{t=0}^{\infty} \left[\prod_{i=0}^t \beta_i \right] \pi_t$$

Moreover,

$$v = (I - L)^{-1} \pi$$

Proof: This is immediate from the result on slide 11 with

$$b(X_{t-1}, X_t) = \beta(X_t) \text{ and } h = \pi$$

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Fixed point theory for state-dependent discounting

Soon we turn to dynamic programming with stochastic discounting

We will use

- an extension of Banach's fixed point theorem for “eventual contractions”
- some useful sufficient conditions for “eventual contractions”

We discuss these fixed point results first

Eventual contractions

Fix

1. $U \subset \mathbb{R}^X$
2. norm $\|\cdot\|$ on U
3. self-map T on U

We call T **eventually contracting** on U if $\exists$ a $k \in \mathbb{N}$ such that T^k is contracting on U under $\|\cdot\|$

Theorem. If U is closed and T is eventually contracting, then T is globally stable on U

Ex. Prove the theorem [Hint: Use Banach's theorem]

Example. Consider $Tu = Au + b$ for $b \in \mathbb{R}^X$ and $A \in \mathcal{L}(\mathbb{R}^X)$

We have already studied the stability properties of T on $\mathbb{R}^X$

We saw in Ch. 1 that $\rho(A) < 1$ implies

1. T is globally stable on $\mathbb{R}^X$
2. The unique fixed point is

$$u^* = (I - A)^{-1}b$$

(The last point follows from $u^* = Au^* + b$ and the NSL)

As an exercise, let's now prove that T is an eventual contraction

Ex. Fixing $u, v \in \mathbb{R}^X$, show by induction that

$$T^k u - T^k v = A^k u - A^k v \quad \text{for all } k \in \mathbb{N}$$

As a result,

$$\|T^k u - T^k v\| = \|A^k u - A^k v\| = \|A^k(u - v)\| \leq \|A^k\| \|u - v\|$$

If $\rho(A) < 1$, we can choose $k \in \mathbb{N}$ such that $\|A^k\| < 1$

(this follows from Gelfand's formula)

$\therefore T$ is eventually contracting on $\mathbb{R}^X$

Note this gives another proof that T is globally stable on $\mathbb{R}^X$

The last example shows the connection to the Neumann series lemma

- adds global stability

But eventual contractions have much wider scope than the Neumann series lemma

- can also be applied in nonlinear settings

A Spectral Radius Condition

Let T be a self-map on $U \subset \mathbb{R}^X$

Proposition. If $\exists$ a positive $L \in \mathcal{L}(\mathbb{R}^X)$ with $\rho(L) < 1$ and

$$|Tv - Tw| \leq L|v - w| \quad \text{for all } v, w \in U$$

then T is an eventual contraction on U

Proof: Fixing $v, w \in U$ and $k \in \mathbb{N}$, we have

$$|T^k v - T^k w| \leq L|T^{k-1} v - T^{k-1} w|$$

or

$$e_k \leq L e_{k-1} \quad \text{where} \quad e_k := |T^k v - T^k w|$$

Ex. Show that $e_k \leq L^k e_0$ for all $k \in \mathbb{N}$

Proof: We know that

$$e_k \leq L e_{k-1} \quad \text{for all } k \in \mathbb{N} \quad (1)$$

We prove by induction

The claim is true at $k = 1$ by (1)

Now suppose that it is true at k , so $e_k \leq L^k e_0$

Since L is order-preserving on U and (1) holds, we have

$$e_{k+1} \leq L e_k \leq L L^k e_0 = L^{k+1} e_0$$

Hence the claim is also true at $k + 1$ — QED

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We have verified $e_k \leq L^k e_0$, or

$$|T^k v - T^k w| \leq L^k |v - w|$$

Let $\|\cdot\|$ be the Euclidean norm

Since $0 \leq a \leq b$ implies $\|a\| \leq \|b\|$, we get

$$\|T^k v - T^k w\| \leq \|L^k |v - w|\| \leq \|L^k\| \|v - w\|$$

Since $\rho(L) < 1$, we have $\|L^k\| \rightarrow 0$ as $k \rightarrow \infty$

Hence T is eventually contracting on U

Blackwell for eventual contractions

In Ch. 2 we studied Blackwell's condition for a contraction

Here we provide an analogous result for eventual contractions

Let $U \subset \mathbb{R}^X$ be such that $v, c \in U$ and $c \geq 0$ implies $v + c \in U$

Proposition. Let T be an order-preserving self-map on U

If $\exists$ a positive $L \in \mathcal{L}(\mathbb{R}^X)$ such that $\rho(L) < 1$ and

$$T(v + c) \leq Tv + Lc \quad \text{for all } c, v \in \mathbb{R}^X \text{ with } c \geq 0$$

then T is eventually contracting on U

Proof: Let U, T, L be as in the statement of the proposition

Fix $v, w \in U$

By the assumed properties on T , we have

$$Tv = T(v + w - w) \leq T(w + |v - w|) \leq Tw + L|v - w|$$

Rearranging gives $Tv - Tw \leq L|v - w|$

Reversing the roles of v and w yields $|Tv - Tw| \leq L|v - w|$

The claim now follows from the proposition on slide 23

MDPs with State-Dependent Discounting

We begin with an MDP $\mathcal{M} = (\Gamma, \beta, r, P)$ with state space X , action space A and feasible state-action pairs G

We replace the constant β with a function β from $G \times X$ to $\mathbb{R}_+$

A function $v \in \mathbb{R}^X$ is said to satisfy the **Bellman equation** if

$$v(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \sum_{x'} v(x') \beta(x, a, x') P(x, a, x') \right\}$$

for all $x \in X$

- discount process can depend on x, a, x'

Possible assumption: $\exists b < 1$ such that

$$\beta(x, a, x') \leq b \text{ for all } (x, a, x') \in G \times X$$

Then

- policy and Bellman operator will be contraction maps
- MDP optimality results easily extend

Unfortunately, this assumption is too strict for many applications

Example. Consider a firm problem where $\beta_t = 1/(1 + r_t)$

Note

$$\beta_t < 1 \implies r_t > 0$$

This is a problem if we want to discount with the real interest rate

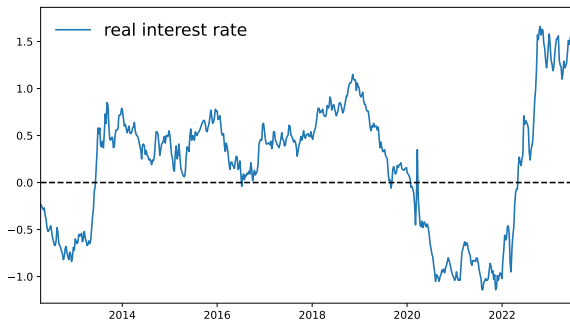


Figure: Real US interest rates are sometimes negative

Also, household preferences are sometimes assumed to have occasionally negative discount rates

- implies β_t sometimes > 1

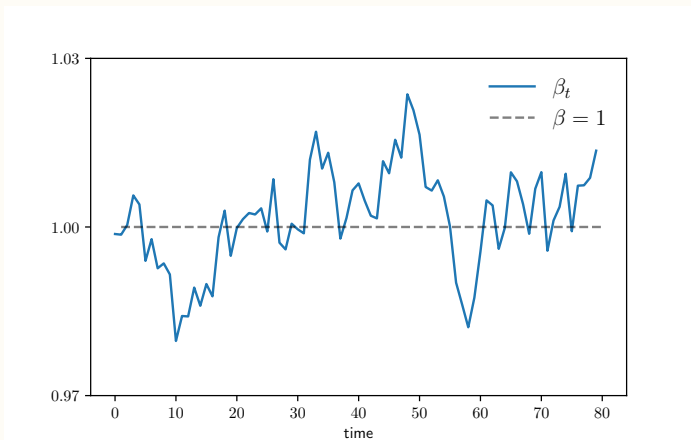


Figure: $(\beta)_{t \geq 0}$ process in Hills, Nakata and Schmidt (2019)

Lifetime values

Let $\Sigma =$ all **feasible policies** (as for regular MDPs)

Given $\sigma \in \Sigma$, the **policy operator** $T_\sigma: \mathbb{R}^X \rightarrow \mathbb{R}^X$ is

$$(T_\sigma v)(x) = r(x, \sigma(x)) + \sum_{x'} v(x') \beta(x, \sigma(x), x') P(x, \sigma(x), x')$$

Set $r_\sigma(x) = r(x, \sigma(x))$ and

$$L_\sigma(x, x') := \beta(x, \sigma(x), x') P(x, \sigma(x), x')$$

Then

$$T_\sigma v = r_\sigma + L_\sigma v$$

Assumption SD. For all $\sigma \in \Sigma$ we have $\rho(L_\sigma) < 1$

Ex. Prove: Assumption SD $\implies T_\sigma$ is globally stable on $\mathbb{R}^X$ with unique fixed point

$$v_\sigma = (I - L_\sigma)^{-1} r_\sigma$$

Proof: Since $T_\sigma v = r_\sigma + L_\sigma v$ we just need to show $\rho(L_\sigma) < 1$

- Then apply the result on slide 20

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The next exercise helps us interpret v_σ

Ex. Let

- $P_\sigma(x, x') = P(x, \sigma(x), x')$
- (X_t) be P_σ -Markov
- $\beta_t = \beta(X_t, \sigma(X_t), X_{t+1})$ for all $t \in \mathbb{N}$ with $\beta_0 = 1$

Prove that, under Assumption SD, the function v_σ obeys

$$v_\sigma(x) = \mathbb{E}_x \sum_{t=0}^{\infty} \left[\prod_{i=0}^t \beta_i \right] r_\sigma(X_t)$$

for all $x \in X$

Proof: Let

$$v(x) := \mathbb{E}_x \sum_{t=0}^{\infty} \left[\prod_{i=0}^t \beta_i \right] r_{\sigma}(X_t)$$

Here we have

- $L_{\sigma}(x, x') = \beta(x, \sigma(x), x')P(x, \sigma(x), x')$ with $\rho(L) < 1$
- (X_t) is P_{σ} -Markov
- $\beta_t = \beta(X_t, \sigma(X_t), X_{t+1})$ for all $t \in \mathbb{N}$ with $\beta_0 = 1$

By the result on slide 11 we have $v = (I - L_{\sigma})^{-1}r_{\sigma}$

Hence $v = v_{\sigma}$, as was to be shown

Greedy policies

The definition is analogous to the MDP case:

Given $v \in \mathbb{R}^X$, a policy σ is called **v -greedy** if, for all x in X ,

$$\sigma(x) \in \operatorname{argmax}_{a \in \Gamma(x)} \left\{ r(x, a) + \sum_{x'} v(x') \beta(x, a, x') P(x, a, x') \right\}$$

Ex. Show that σ is v -greedy whenever $T_\sigma v = Tv$

Proof: Similar to the MDP case

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Proof: Similar to the MDP case

Optimality

Let Assumption SD hold

Analogous to the MDP case, we define the **value function** via

$$v^* := \bigvee_{\sigma \in \Sigma} v_{\sigma}$$

A policy σ is called **optimal** if $v_{\sigma} = v^*$

The **Bellman operator** $T: \mathbb{R}^X \rightarrow \mathbb{R}^X$ takes the form

$$(Tv)(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \sum_{x'} v(x') \beta(x, a, x') P(x, a, x') \right\}$$

Here is our main optimality result for MDPs with state-dependent discounting

Proposition. If Assumption SD holds, then

1. v^* is the unique fixed point of the Bellman operator,
2. a policy $\sigma \in \Sigma$ is optimal if and only if it is v^* -greedy, and
3. at least one optimal policy exists

We state and prove a more general result later

- see Ch. 8

For now let's

- Consider algorithms
- Look at a special case, where (β_t) is purely exogenous
- Examine some applications

Algorithms for state-dependent discounting MDPs

For MDPs we studied

- value function iteration (VFI)
- Howard policy iteration (HPI)
- optimistic policy iteration (OPI)

Do these algorithms still converge?

Under what conditions?

The statement of the VFI and OPI algorithms are identical

- use the new definitions for T, T_σ

The statement of HPI is as follows

Algorithm 1: HPI for MDPs with state-dependent discounting

input $\sigma \in \Sigma$

$v_0 \leftarrow v_\sigma$ and $k \leftarrow 0$

repeat

$\sigma_k \leftarrow$ a v_k -greedy policy

$v_{k+1} \leftarrow (I - L_{\sigma_k})^{-1} r_{\sigma_k}$

if $v_{k+1} = v_k$ **then break**

$k \leftarrow k + 1$

return σ_k

Theorem. Under Assumption SD the following statements hold

1. If (v_k) is generated by OPI / VFI, then

$$v_k \rightarrow v^* \quad (k \rightarrow \infty)$$

2. HPI returns an optimal policy in finitely many steps

These convergence results are special cases of results in Ch. 8

Special case: exogenous discounting

Consider a model (Γ, β, r, Q, R) where

1. Γ is a nonempty correspondence from $Y \rightarrow A$; set

$$G := \{(y, a) \in Y \times A : a \in \Gamma(y)\}$$

2. β is a function from Z to $\mathbb{R}_+$
3. r is a function from G to $\mathbb{R}$
4. Q is a stochastic matrix on Z
5. R is a stochastic kernel from G to Y

A summary of dynamics:

- (Z_t) is Q -Markov
- The **discount factor process** $(\beta_t)_{t \geq 0}$ obeys $\beta_t := \beta(Z_t)$
- Given $Y_t = y$ and current action a , current reward is $r(y, a)$
- Y_{t+1} is drawn from distribution $R(y, a, \cdot)$
- Y_{t+1} and Z_{t+1} are updated independently given (y, z, a)

The Bellman equation becomes

$$v(y, z) = \max_{a \in \Gamma(y)} \left\{ r(y, a) + \beta(z) \sum_{z', y'} v(y', z') Q(z, z') R(y, a, y') \right\}$$

for all $(y, z) \in X$

Given $\sigma \in \Sigma$, the **policy operator** is

$$(T_\sigma v)(y, z) = r(y, \sigma(y, z)) + \beta(z) \sum_{z', y'} v(y', z') Q(z, z') R(y, \sigma(y, z), y')$$

Proposition Let

$$L(z, z') := \beta(z)Q(z, z') \quad (2)$$

If $\rho(L) < 1$, then all of the optimality results for MDPs with state-dependent discounting on slide 40 apply

See the book for a proof

How strict is the condition $\rho(L) < 1$?

Let's recall the discount factor process in Hills et al (2019)

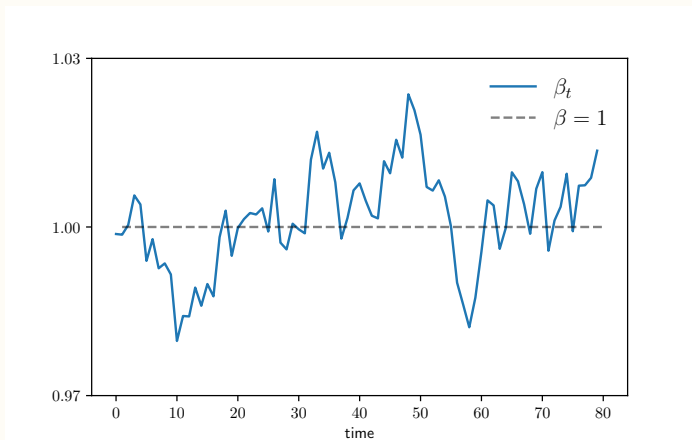


Figure: $(\beta)_{t \geq 0}$ process in Hills, Nakata and Schmidt (2019)

Calculating the spectral radius at the parameters of Hills et al gives

$$\rho(L) = 0.9996$$

Hence

- Assumption SD holds
- Optimality results apply
- VFI, OPI, HPI converge

Application: Inventory Management

Recall the inventory management model with Bellman equation

$$v(x) = \max_{a \in \Gamma(x)} \left\{ r(x, a) + \beta \sum_{d \geq 0} v(f(x, a, d)) \varphi(d) \right\}$$

- $x \in X := \{0, \dots, K\}$ is the current inventory level
- a is the current inventory order
- $r(x, a)$ is current profits
- $f(x, a, d) := (x - d) \vee 0 + a$
- d is an IID demand shock with distribution φ

We now replace β with $\beta_t = \beta(Z_t)$

- $(Z_t)_{t \geq 0}$ is Q -Markov on Z

This is an MDP with state-dependent discounting

The Bellman equation becomes

$$v(y, z) = \max_{a \in \Gamma(y)} \left\{ r(y, a) + \beta(z) \sum_{d, z'} v(f(y, a, d), z') \varphi(d) Q(z, z') \right\}$$

All optimality results hold when

- $L(z, z') := \beta(z)Q(z, z')$ and
- $\rho(L) < 1$

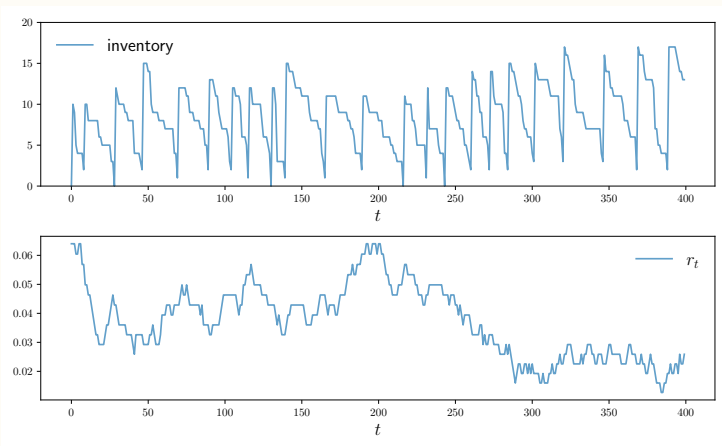


Figure: Inventory dynamics with time-varying interest rates

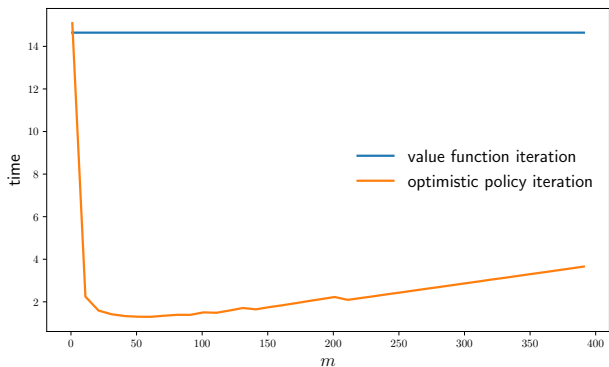


Figure: OPI vs VFI timings for the inventory model

Introduction to asset pricing

We end this chapter with a brief discussion of asset pricing

- Related to calculations with state-dependent discounting
- Readers who wish to focus on dynamic programming can jump to the next chapter

In standard neoclassical asset pricing, state-dependent discounting arises naturally from basic assumptions

- prices are determined by market equilibria
- no arbitrage

Risk-neutral pricing

Consider an asset with payoff G_{t+1} next period

What current price Π_t should we assign?

Risk neutral pricing says

$$\Pi_t = \mathbb{E}_t \beta G_{t+1}$$

for some $\beta \in (0, 1)$

If the payoff is in k periods, then the price is $\mathbb{E}_t \beta^k G_{t+k}$

Example. The time t risk-neutral price of a **European call option** is

$$\Pi_t = \mathbb{E}_t \beta^k \max\{S_{t+k} - K, 0\}$$

where

- S_t is the price of the underlying asset (e.g., stock)
- K is the strike price
- k is the duration
- $\beta = 1/(1 + r)$ where r is the discount rate

But assuming risk neutrality for all investors is **not consistent with the data**

Example. Consider the rate of return $r_{t+1} := (G_{t+1} - \Pi_t)/\Pi_t$

From $\Pi_t = \mathbb{E}_t \beta G_{t+1}$ we get

$$\mathbb{E}_t \beta \frac{G_{t+1}}{\Pi_t} = 1 \quad \Longleftrightarrow \quad \mathbb{E}_t \beta (1 + r_{t+1}) = 1$$

Hence

$$\mathbb{E}_t r_{t+1} = \frac{1 - \beta}{\beta}$$

Thus, risk neutrality implies that all assets have the same expected rate of return

In fact riskier assets usually have higher average rates of return

- incentivize investors to bear risk

Example. The **risk premium** $:=$ expected rate of return minus the rate of return on a risk-free asset

Risk-neutrality $\implies$ risk premium is zero for all assets

But calculations based on post-war US data show that

average risk premium for equities $\approx 8\%$ per annum

These facts motivate a more general theory...

Fundamental theorem of asset pricing

Here is an informal statement from standard neoclassical finance
— Stephen Ross, LP Hansen, David Kreps, etc.

There exists a positive random variable M_{t+1} such that the price Π_t of any payoff G_{t+1} obeys

$$\Pi_t = \mathbb{E}_t M_{t+1} G_{t+1}$$

- assumptions $\approx$ representative agent, no arbitrage
- M_{t+1} is called the **stochastic discount factor** (SDF)
- can handle risk aversion, different rates of return
- key assertion: same SDF can price any asset

Special case: two period Lucas tree

How should M_{t+1} be constructed?

To answer this we need a model

Let's look at a simple example

- a two-period model
- CRRA utility
- one asset and one agent

Agent takes Π_t as given and solves

$$\max_{0 \leq \alpha \leq 1} \{u(C_t) + \beta \mathbb{E}_t u(C_{t+1})\}$$

subject to $C_t = E_t - \Pi_t \alpha$ and $C_{t+1} = E_{t+1} + \alpha G_{t+1}$

Here

- u is a flow utility function and β measures impatience
- G_{t+1} is the payoff of the asset and Π_t is the time- t price
- E_t and E_{t+1} are endowments and
- α is the share of the asset purchased by the agent

Rewrite as

$$\max_{\alpha} \{u(E_t - \Pi_t \alpha) + \beta \mathbb{E}_t u(E_{t+1} + \alpha G_{t+1})\}$$

Differentiating w.r.t. α leads to first order condition

$$u'(E_t - \Pi_t \alpha) \Pi_t = \beta \mathbb{E}_t u'(E_{t+1} + \alpha G_{t+1}) G_{t+1}$$

Rearranging gives us

$$\Pi_t = \mathbb{E}_t M_{t+1} G_{t+1} \quad \text{where} \quad M_{t+1} := \beta \frac{u'(C_{t+1})}{u'(C_t)}$$

Example. In the CRRA case $u(c) = c^{1-\gamma}/(1-\gamma)$ we get

$$M_{t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma}$$

Alternatively,

$$M_{t+1} = \beta \exp(-\gamma g_{t+1}) \quad \text{where} \quad g_{t+1} := \ln(C_{t+1}/C_t)$$

Applies

- heavier discounting in states of the world where consumption growth is high
- lower discounting in states of the world where consumption growth is low

Favors assets that hedge against the risk of low consumption states

Question: How well does this model work when confronted with data?

Answer: badly — search “equity premium puzzle” for an introduction

1. Shiller (1982), Mehra and Prescott (1985), etc.

Recent quantitative models build more sophisticated SDFs to try to get closer to the data

- Epstein–Zin preferences
- ambiguity
- long-run risk models, etc.

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General SDF, Markov state

Let's go back to the general case

$$\Pi_t = \mathbb{E}_t M_{t+1} G_{t+1}$$

How can we compute this?

Suppose $(X_t)_{t \geq 0}$ is P -Markov on X ,

$$M_{t+1} = m(X_t, X_{t+1}) \quad \text{and} \quad G_{t+1} = g(X_t, X_{t+1})$$

With $\pi(x) := \mathbb{E}_x M_{t+1} G_{t+1}$, we have

$$\pi(x) = \sum_{x' \in X} m(x, x') g(x, x') P(x, x')$$

Pricing a dividend stream

Consider the price of a claim on dividend stream $(D_t)_{t \geq 0}$

Let the price at time t be Π_t

Buying at t and selling at $t + 1$ pays $\Pi_{t+1} + D_{t+1}$

Hence the price sequence $(\Pi_t)_{t \geq 0}$ must obey

$$\Pi_t = \mathbb{E}_t M_{t+1} (\Pi_{t+1} + D_{t+1})$$

Current price depends on future price — how can we solve it?

Recall the key equation

$$\Pi_t = \mathbb{E}_t M_{t+1}(\Pi_{t+1} + D_{t+1})$$

Let

- $D_t = d(X_t)$ where $(X_t)_{t \geq 0}$ is P -Markov
- $\pi(x)$ = current price given $X_t = x$

We get

$$\pi(x) = \sum_{x'} m(x, x')(\pi(x') + d(x'))P(x, x') \quad (x \in \mathbf{X})$$

Rewrite the last expression as

$$\pi = A\pi + Ad$$

where

$$A(x, x') := m(x, x')P(x, x')$$

Neumann series lemma: $\rho(A) < 1 \implies$ the unique solution is

$$\pi^* = (I - A)^{-1}Ad$$

- π^* is called an **equilibrium price function**
- A is called the **Arrow–Debreu discount operator**

Forward Sum Representation

Asset prices can also be written as infinite sums

For a **cum dividend** contract the price vector obeys

$$\pi = d + A\pi \quad \text{with} \quad A(x, x') := m(x, x')P(x, x')$$

When $\rho(A) < 1$, the Neumann series lemma gives

$$\pi(x) = \mathbb{E}_x \sum_{t=0}^{\infty} \left[\prod_{i=0}^t M_i \right] D_t \quad (x \in \mathbb{X})$$

where $M_{t+1} := m(X_t, X_{t+1})$ and $M_0 := 1$

Price = EPV of dividends, with D_t discounted by $M_1 \cdots M_t$

Nonstationary Dividends

A more realistic model is one where dividends grow over time

A standard model of dividend growth is

$$\ln \frac{D_{t+1}}{D_t} = \kappa(X_t, \eta_{t+1}) \quad t = 0, 1, \dots,$$

Here

- κ is a fixed function
- (X_t) is P -Markov on finite set X
- (η_t) is IID with density φ
- $M_{t+1} = m(X_t, X_{t+1})$ for some positive function m

Growing dividends $\implies$ growing prices

- no π such that $\Pi_t = \pi(X_t)$ for all t

Instead we try to solve for the **price-dividend ratio** $V_t := \Pi_t/D_t$

Ex. Show that $\Pi_t = \mathbb{E}_t [M_{t+1}(D_{t+1} + \Pi_{t+1})]$ implies

$$V_t = \mathbb{E}_t [M_{t+1} \exp(\kappa(X_t, \eta_{t+1})) (1 + V_{t+1})]$$

Conditioning on $X_t = x$,

$$v(x) = \sum_{x' \in X} m(x, x') \int \exp(\kappa(x, \eta)) \varphi(d\eta) [1 + v(x')] P(x, x')$$

Let

$$A(x, x') := m(x, x') \int \exp(\kappa(x, \eta)) \varphi(d\eta) P(x, x')$$

Now we see a v that solves

$$v(x) = \sum_{x' \in X} [1 + v(x')] A(x, x')$$

Equivalent:

$$v = A\mathbb{1} + Av$$

If $\rho(A) < 1$, then the unique solution is

$$v^* = (I - A)^{-1} A\mathbb{1}$$

Example. Dividend growth is

$$\kappa(X_t, \eta_{d,t+1}) = \mu_d + X_t + \sigma_d \eta_{d,t+1} \quad \text{where} \quad (\eta_{d,t})_{t \geq 0} \stackrel{\text{iid}}{\sim} N(0, 1)$$

Consumption growth is given by

$$\ln \frac{C_{t+1}}{C_t} = \mu_c + X_t + \sigma_c \eta_{c,t+1} \quad \text{where} \quad (\eta_{c,t})_{t \geq 0} \stackrel{\text{iid}}{\sim} N(0, 1)$$

We use the Lucas CRRA SDF, implying that

$$M_{t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\gamma} = \beta \exp(-\gamma(\mu_c + X_t + \sigma_c \eta_{c,t+1}))$$

```
using QuantEcon, LinearAlgebra
```

```
"Creates an instance of the asset pricing model with Markov state."
```

```
function create_asset_pricing_model(;  
    n=200,                # state grid size  
    ρ=0.9, v=0.2,          # state persistence and volatility  
    β=0.99, γ=2.5,        # discount and preference parameter  
    μ_c=0.01, σ_c=0.02,    # consumption growth mean and volatility  
    μ_d=0.02, σ_d=0.1)    # dividend growth mean and volatility  
    mc = tauchen(n, ρ, v)  
    x_vals, P = exp.(mc.state_values), mc.p  
    return (; x_vals, P, β, γ, μ_c, σ_c, μ_d, σ_d)  
end
```

```

" Build the discount matrix A. "
function build_discount_matrix(model)
    (; x_vals, P,  $\beta$ ,  $\gamma$ ,  $\mu_c$ ,  $\sigma_c$ ,  $\mu_d$ ,  $\sigma_d$ ) = model
    e = exp. ( $\mu_d - \gamma\mu_c + (\gamma^2\sigma_c^2 + \sigma_d^2)/2$  .+  $(1-\gamma)*x\_vals$ )
    return  $\beta * e .* P$ 
end

"Compute the price-dividend ratio associated with the model."
function pd_ratio(model)
    (; x_vals, P,  $\beta$ ,  $\gamma$ ,  $\mu_c$ ,  $\sigma_c$ ,  $\mu_d$ ,  $\sigma_d$ ) = model
    A = build_discount_matrix(model)
    @assert maximum(abs.(eigvals(A))) < 1 "Requires  $r(A) < 1$ ."
    n = length(x_vals)
    return (I - A) \ (A * ones(n))
end

```

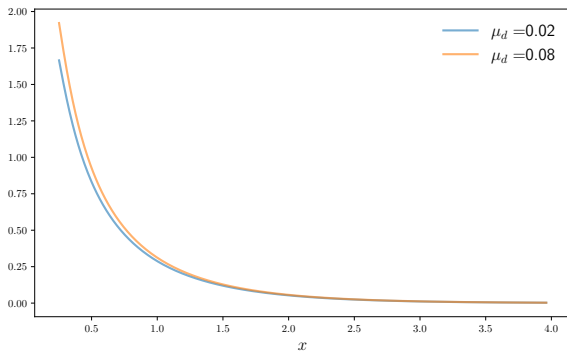


Figure: Price-dividend ratio as a function of x

Incomplete Markets

So far the Neumann series lemma solved for the equilibrium price vector π

But some modifications introduce **nonlinearities** that render it inapplicable

Example. Harrison and Kreps (1978): heterogeneous beliefs and incomplete markets

- two types of agents
- type i believes the state updates via stochastic matrix P_i , for $i = 1, 2$
- agents are risk-neutral, so $m(x, y) \equiv \beta \in (0, 1)$

For their model the equilibrium condition becomes

$$\pi(x) = \max_i \beta \sum_{x'} [\pi(x') + d(x')] P_i(x, x') \quad (x \in \mathbf{X})$$

The max makes this **nonlinear** — so we turn it into a fixed point problem

Define $T: \mathbb{R}_+^{\mathbf{X}} \rightarrow \mathbb{R}_+^{\mathbf{X}}$ by

$$(T\pi)(x) = \max_i \beta \sum_{x'} [\pi(x') + d(x')] P_i(x, x')$$

With $d \geq 0$, T is a self-map on $\mathbb{R}_+^{\mathbf{X}}$, and π solves the equilibrium condition iff $T\pi = \pi$

Claim. T is a contraction of modulus β on $\mathbb{R}_+^X$ under $\|\cdot\|_\infty$

Proof: Fix $p, q \in \mathbb{R}_+^X$. Using the elementary bound $|\vee_i a_i - \vee_i b_i| \leq \max_i |a_i - b_i|$ and the triangle inequality,

$$\begin{aligned} |(Tp)(x) - (Tq)(x)| &\leq \beta \max_i \sum_{x'} |p(x') - q(x')| P_i(x, x') \\ &\leq \beta \|p - q\|_\infty \end{aligned}$$

Taking the max over x gives $\|Tp - Tq\|_\infty \leq \beta \|p - q\|_\infty$ □

$\mathbb{R}_+^X$ is closed, so T has a unique fixed point π^* — the equilibrium prices — computable by successive approximation

Ex. Reprove contractivity using Blackwell's condition